

Student Performance Tracker Using Machine Learning

1. Abstract

This project is about predicting student performance using Machine Learning. The system analyzes weekly self-study hours, attendance percentage, and class participation to predict the final score. Linear Regression algorithm is used for prediction. The project helps in understanding how machine learning can be used in education.

2. Introduction

Student performance prediction is an important educational problem that helps identify academic progress and learning patterns. Machine learning helps automate the prediction process by analyzing student data and identifying meaningful patterns.

Objectives

- To learn machine learning basics.
- To predict student performance using academic data.
- To understand Linear Regression.
- To analyze factors affecting student scores.

3. Problem Statement

Student performance depends on factors such as study hours, attendance, and participation. This project uses machine learning to predict student scores based on academic behavior.

4. Scope of the Project

- Loading the dataset
- Data visualization
- Training the machine learning model
- Predicting results
- Checking model accuracy

5. Technologies Used

Programming Language

Python

Libraries Used

Pandas

Matplotlib

Seaborn

Scikit-learn

Development Environment

Jupyter Notebook / Google Colab

6. Dataset Description

The project uses a Student Performance Dataset.

Dataset Features

- student_id – Unique student identifier
- weekly_self_study_hours – Average weekly study hours
- attendance_percentage – Attendance percentage
- class_participation – Classroom participation score
- total_score – Final performance score
- grade – Grade category

Target Variable

- total_score

7. System Architecture

1. Data Collection
2. Data Preprocessing
3. Feature Selection
4. Train-Test Split
5. Model Training
6. Prediction
7. Performance Evaluation

8. Methodology

Step 1: Load Dataset

Step 2: Data Visualization

Step 3: Split Dataset

Step 4: Train Model using Linear Regression

Step 5: Predict Student Score

Step 6: Evaluate using R^2 Score

9. Machine Learning Model Used

Linear Regression

Linear Regression is a supervised machine learning algorithm used to predict numerical values. In this project, it predicts student total scores.

10. Advantages

- Simple and efficient algorithm
- Easy to interpret
- Suitable for prediction problems • Beginner-friendly

11. Limitations

- Depends on dataset quality
- Limited features may affect accuracy
- Results depend on training data

12.Conclusion

The Student Performance Tracker provides an effective solution for predicting student scores based on study habits, attendance, and participation. It demonstrates the complete machine learning workflow from data analysis to prediction.